



CYBER SECURITY

This document covers cybersecurity, hacking, notable cyberattacks, HTTP vs. HTTPS, and basic explanations of AES

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Cybersecurity: Keeping Our Digital World Safe

Cybersecurity means keeping computers, networks, and data safe from people who try to steal information or cause harm.

Why it Matters: As we use the internet for more things, it's important to protect our personal details, money, and privacy from online threats.

Cybersecurity vs. Hacking: Two Sides of the Coin

- **Cybersecurity:** The practice of protecting systems and data from attacks, ensuring they remain secure.
 - **Hacking:** The act of breaking into systems, sometimes illegally, to access or disrupt information. **Ethical hackers** use their skills to help improve security.
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Famous Cyber Attacks: What Happened?

1. **WannaCry Ransomware (2017):**
 - **What Happened:** A virus spread through a flaw in Windows, locking files and asking for money to unlock them.
 2. **Equifax Breach (2017):**
 - **What Happened:** Hackers found a weakness in a website tool and stole personal info from millions of people.
 3. **Target Breach (2013):**
 - **What Happened:** Hackers got into Target's system through a third-party vendor, stealing credit card details.
 4. **Yahoo Breach (2013-2014):**
 - **What Happened:** Hackers used phishing and weak security to access data from 3 billion accounts.
 5. **SolarWinds Attack (2020):**
 - **What Happened:** Malicious software was hidden in updates, affecting many companies and government systems.
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HTTP vs. HTTPS: What's the Difference?

- **HTTP:** Sends data over the web without encryption, making it easier to intercept.

- **HTTPS:** Adds a layer of security by encrypting the data, making it safer for users.

Encryption: How AES and RSA Keep Us Safe

AES – One Key for Both:

- **Type:** Symmetric encryption, using the same key to lock and unlock data.
- **Example:** Like sharing a password to open a locked box.

RSA – Two Keys for Security:

- **Type:** Asymmetric encryption, using a public key to lock and a private key to unlock.
- **Example:** Like sending a locked box that only the recipient can open with their key.

